Estimating loop length from CryoEM images at medium resolutions

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Abstract

Background

Cryoelectron Microscopy (CryoEM) is an important biophysical technique that produces 3-dimensional (3D) images at different resolutions. De novo modeling is becoming a promising approach to derive the atomic structure of proteins from CryoEM 3D images at medium resolutions. Distance measurement along a 1-dimensional (1D) skeleton in the 3D image is an important step in de novo modeling. The points on the skeleton often need to be processed to accurately estimate the distance that corresponds to the loop length. In spite of the need of such measurement, little has been investigated about the accuracy of the measurement in searching for an effective method.

Results

We have developed a new computational geometric approach to derive a simplified curve in order to estimate the loop length along the skeleton. The method was tested using fifty simulated density images of helix-loop-helix segments of atomic structures and eighteen experimentally derived density data from Electron Microscopy Data Bank (EMDB). The test using simulated density maps shows that it is possible to estimate within 0.5$\mathrm{\AA}$ difference of the expected length for 48 of the 50 cases. The test of eighteen experimentally derived CryoEM images shows

that twelve cases have error within 2Å.

Conclusions

The tests using both simulated and experimentally derived images show that it is possible for our proposed method to reasonably estimate the loop length along the skeleton if the secondary structure elements, such as α-helices, can be detected accurately.

Keywords

Protein, Bioinformatics, Computational Geometry, Cryoelectron Microscopy, Topology, Secondary Structure, Loop

Background

Cryoelectron microscopy (CryoEM) continues to produce 3-dimensional (3D) images of large protein complexes. The Electron Microscopy Data Bank (EMDB) currently archives the 3D images, referred to as density maps in this paper, with a wide range of resolutions from 3Å to over 80Å [1]. When the density map is resolved to high resolution (3-5Å) such as in recent EMDB deposits 5499, 5520, 5495 and 5496 [2,3], it is possible to derive the near atomic structure from the density map. However, when the density map is not resolved to the high resolution range, it is still challenging to derive the structure of the imaged molecule [4,5]. Fitting and comparative modeling approaches have been developed to utilize the existing atomic structures in the Protein Data Bank (PDB). These approaches apply when a component of the target density map has been resolved to near atomic resolution structure or when the target protein shares significant homology with existing atomic structures [6–9].

De novo modeling has been demonstrated as a promising method to derive atomic protein structures from 3D images at medium resolution, such as 5-10Å [10–13], without relying on the availability of a template structure such as those found in PDB. The input includes the protein density map (top left in Figure 1), and the amino acid sequence of the protein (top right of Figure 1). First, the secondary structure elements (SSEs) such as α-helices (red sticks in Figure 1) and β-sheets are often identified using pattern recognition methods [14–19]. Skeletonization methods detect the underlying 1D structure (green, left in Figure 1) of a 3D image's iso-surface [13, 20]. Next, the amino acid sequence segments (red cylinders, right of Figure 1) of the SSEs can be predicted using existing prediction tools [21–24]. The topology of the secondary structures can then be inferred by combining the two sources of information about the secondary structures, one from

the 3D image and the other from the protein sequence [11, 12, 25–27]. Once the topology of the SSEs is determined, the atomic structure can be modeled [11, 12, 16].

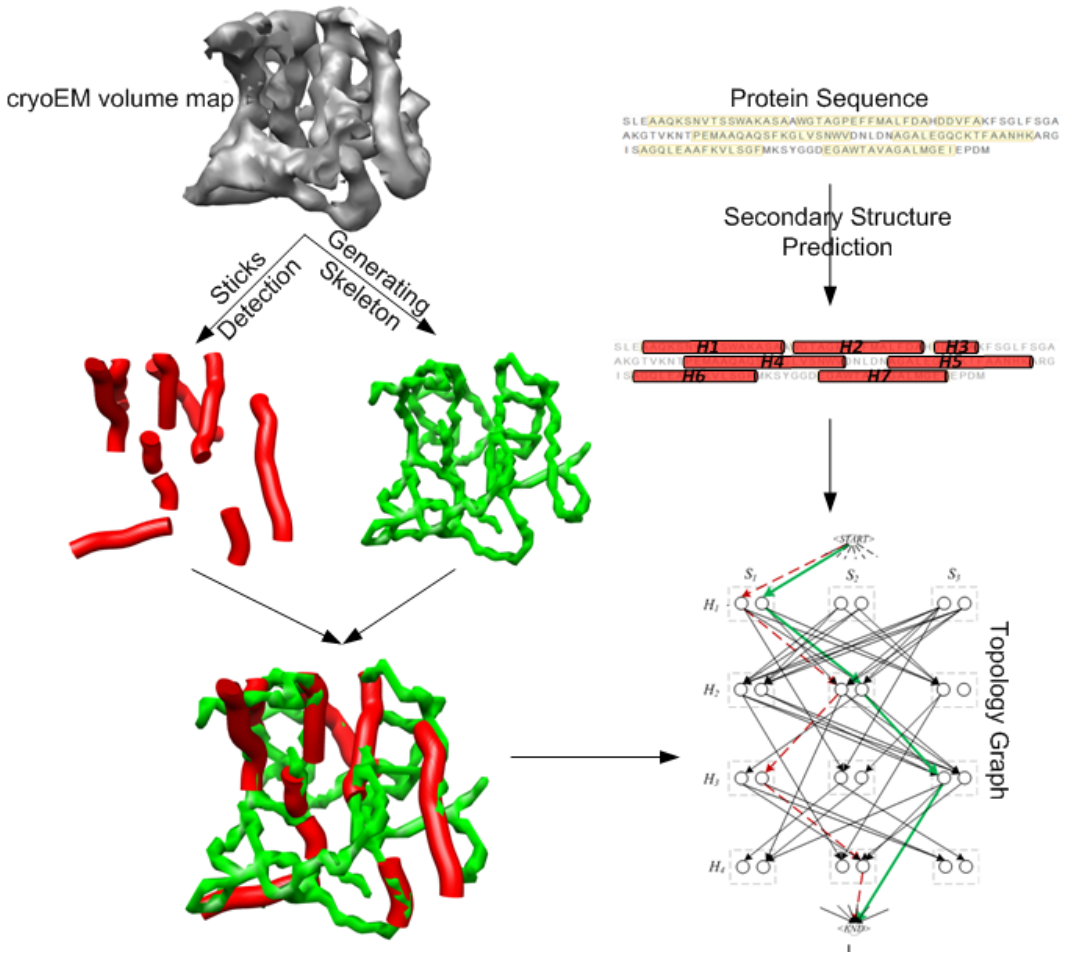


Figure 1: Deriving the topology of the secondary structure elements from CryoEM images. The skeleton (green) and detected helices (red) derived from the density map (gray) are combined with the predicted sequence segments of the helices to form a topology graph [10, 11, 26].

An amino acid sequence has a direction, from the N-terminal to the C-terminal. When the protein sequence is threaded through the 3D image, it visits the detected helix sticks in a unique order. The determination of this order is the topology problem. The native topology of a protein's SSEs is likely to produce the lowest energy state compared to incorrect topologies [28]. The topology of the SSEs is a critical piece of information in protein structure prediction. It has been demonstrated that the topology of the helix sticks can be derived using graph-matching approaches [11, 25, 26]. Our previous work translated the graph-matching problem into a constrained shortest path problem in a topology graph, for which we presented an

$O(N^2 2^N)$ dynamic programming algorithm [11]. We later used a dynamic graph approach to handle the errors in the data [26].

The idea of topology determination is to use the distances between the SSEs as a metric in matching. For example, two helices 5Å apart in the 3D image should be matched to two helices of similar distance in the protein sequence. The distance between two ends of two helices (one on each) can be simply estimated as their Euclidean distance [4], or it can be measured more accurately along the skeleton [10, 26]. The distance between two helices on the amino acid sequence can be translated into the distance in 3D space by assuming a 3.8Å separation between two adjacent amino acids on the sequence. The assumption is that the correct topology results in the overall best match when all pairs of helices are considered.

Skeletonization of the 3D image is a powerful method to extract the connections between the SSEs. The skeleton can be derived using thinning and pruning methods [20]. It roughly represents the major paths in the 3D image. The skeleton is a subset of the density map's voxels that describe such paths. Ideally, the distance between two specific ends of two helices should be measured along the skeleton connecting the two ends. If we simply add the length of the line segments along the path, there is a danger of over estimation due to the high potential for right angles in the path. Moreover, the skeleton is expected to contain errors, since the 3D image often contains errors. It is not clear if the skeleton length estimation methods are accurate enough for topology determination.

This paper introduces a new method to approximate the distances between the detected SSEs from a density map, for use in the topology determination algorithm. In particular, we measure the length along the skeleton using a combination of graph-theoretic and computational geometric methods. We tested the method using both simulated density maps and the experimentally derived density maps from EMDB. The measured length appears to agree with the expected length when the SSEs are detected fairly accurately.

Results and discussion

Test data and overall process

Two data sets were used in testing performance. The simulated data set consists of fifty randomly selected helix-loop-helix (HLH) motifs from atomic structures in PDB. The proteins extracted exhibit less than 10% sequence identity. Each extracted HLH of the protein structure was used to generate a 3D density map using EMAN1.9 pdb2mrc [29]. The density maps were simulated to 8Å resolution.

The real data set consists of 18 cases whose density maps were downloaded from EMDB with resolution from 4.2Å to 6.8Å. Their EMDB entries are 5030 (6.4Å), 1733 (6.8Å), 5001 (4.2Å), 1740 (6.8Å) and 5168

(6.6Å). Each of these density maps is aligned with their PDB structures at download.

The length of a loop was measured along the skeleton voxel points between (and including) the end points of the two surrounding helices. The endpoint of a helix represents the end of the central axis of the helix [14, 15]. The helices were detected using SSETracer, a simplified version of SSELearner [19]. The skeleton was detected using a local maximum clustering method, more details of which are forthcoming in a separate paper. In order to test the accuracy of our algorithm, we visually inspected the detected helices and included only those cases in which the helices were roughly accurate. This was done to isolate the potential error in our loop length estimation and not helix detection.

Accuracy

The accuracy of the measurement was evaluated using both the simulated data and the real data from the EMDB. Table 1 summarizes the results for the simulated data. The input to our method includes two pieces of information: the detected helix (red sticks) end points and the skeleton voxels (red dots) (Figure 2B). Each measured length along the skeleton was compared with the expected length of the loop. The expected length was calculated as 3.8Å $\times(n+1)$, where n is the number of the amino acids on the loop and 3.8Å is the average distance between two amino acids. We add 1 to n because there is always one more interval between amino acids than there are amino acids in the loop. For example, a loop with one amino acid has a gap between each helix endpoint (two total), so with $n = 1$ we have 2 gaps and an expected length of 2 $\times$ 3.8Å.

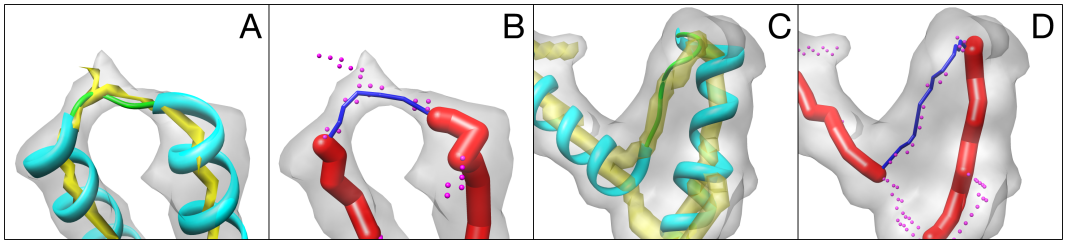


Figure 2: Loop length estimation from a simplified curve. The density map (gray), detected helices (red sticks), the true structure (cyan) are shown for the helix-loop-helix portion of the structure for 1DU0 (PDB Id) in (A, B) and 1MW8 in (C, D). The detected skeleton (yellow) is shown as surface view in (A) and (C), as voxels (red dots) in (B) and (D). The simplified curve derived is shown in blue.

The 50 tested cases were sorted by the length of the loop, ranging from 1 to 10 amino acids. Almost all the 50 test cases appear to have the error within 0.5Å (column 6 of Table 1). As an example, the loop in 1DU0 (row 15 of Table 1) has three amino acids and the expected length of the loop is 15.2Å. The measured

length of the loop along the skeleton is 14.99Å. The relative error is 1.4% of the expected loop length. The simplified curve (blue in Figure 2B) detected by the algorithm appears to be close to the skeleton points (red dots). Another example is from 1MW8 (Figure 2 C, D, row 29 of Table 1) with six amino acids on the loop. The error of the measurement is 0.358Å in this case (column 6 of row 29, Table 1). Note that the skeleton points branch into multiple directions (Figure 2D), yet the algorithm correctly measured the length between the two ending points of the helices by using Hausdorff measurements (see Algorithm). In some cases, as in rows 18 and 28 in Table 1, the greedy step in the Hausdorff computation breaks down and we measure between the wrong pair of endpoints or use the wrong skeleton segment to model the loop length.

The test using the experimentally derived density data involves 18 HLH motifs from density maps with 4-7Å resolution from EMDB. 12 of the 18 cases have measured error within 2Å, and 6 have error between 2Å and 5Å. The real density maps from the experiments are often more challenging with missing density and additional densities that do not align with the true structure. The helices and skeletons detected from the real maps are therefore often less accurate than those from the simulated density maps. Figure 3 shows an example of experimentally derived data in EMDB 5168 (row 15 in Table 2). Its relative error is 15.2%, higher than a comparable case with a synthetic density map used instead. In general, we saw an increase in error using the real density images, due to greater errors in helix detection and skeletonization induced by the noise present.

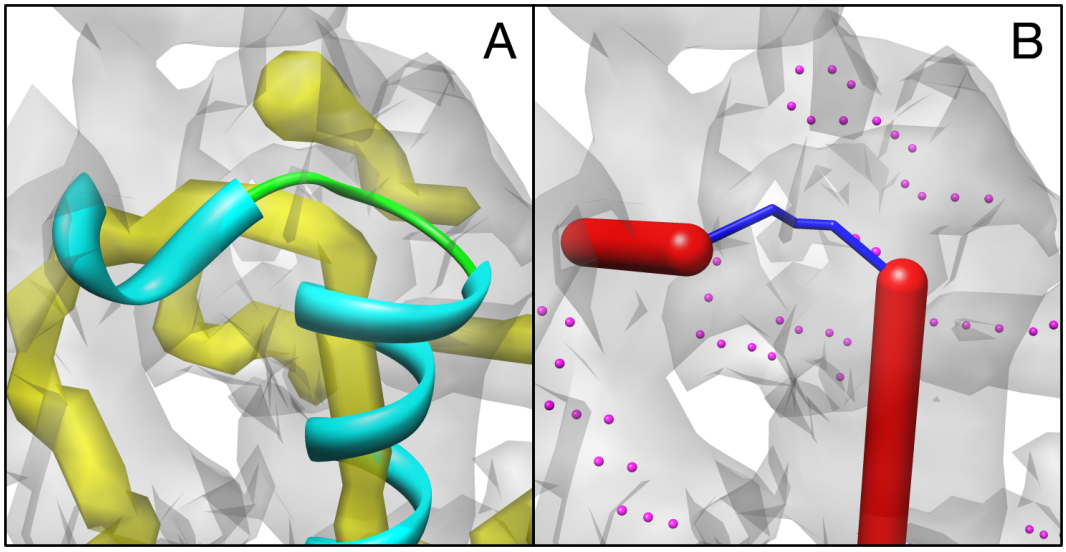


Figure 3: Detected simplified curve for a loop in CryoEM image (EMDB 5168). The color scheme is the same as that in Figure 2.

The algorithm uses a simplification parameter ϵ that is user defined. ϵ is the width of the vertex removal

band (refer to the algorithm and Figure 7 for more details). In general, the smaller the ϵ value, the less change in the simplified curve compared to the initial path. In some cases, the detected helices my be too long, making the estimated distance too short. Hence, we chose $\epsilon = 0$ because no simplification is needed. In other cases, the initial path may be so long that in order to simplify it to the point that its length is close to the expected length we would need a much larger value of ϵ. In order to see the degree of simplification that produced the most accurate results, we sampled ϵ's range inside [0,6] in increments of 0.05. The lengths of the outputs form a step function compared to the respective ϵ values, and the value closest to the expected value can be chosen, as in Figure 4 (left). When a range of values produces identical simplifications, we took the largest ϵ in that range–this is an arbitrary choice that we define for the sake of consistency. When the expected length fell between two steps of the function, we took the lesser distance to a step. In the case of Figure 4 (left), we chose 0.6 instead of 1, because $a < b$, meaning the expected length is closer to a simplification with $\epsilon \in [0, 0.6]$ rather than the one produced by $\epsilon \in (0.6, 1.5]$.

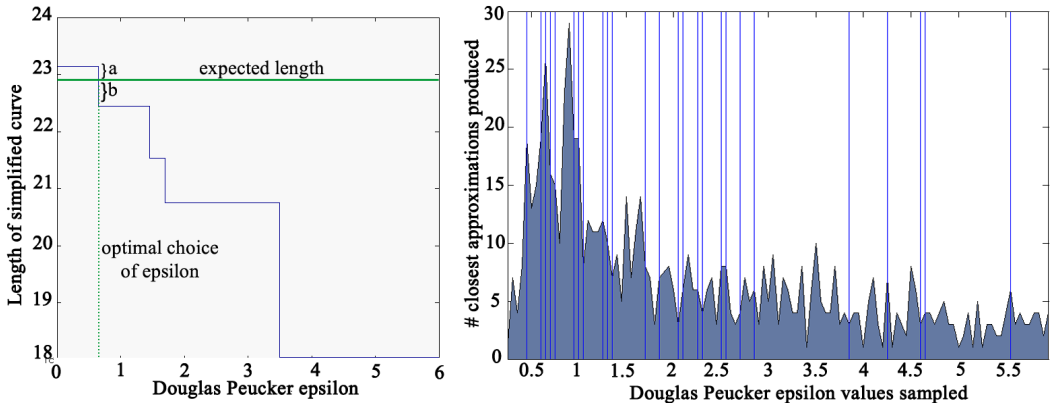


Figure 4: The Douglas-Peucker parameter ϵ. (Left) The ϵ step function for case 21 in Table 1 (PDB 1D8L), with the value of ϵ used for the best estimate. (Right) Distribution of the best ϵ in the simulated data set of 800 loops. The vertical lines show the values that are listed in Table 1.

Figure 4 (right) shows the distribution of the values of ϵ for about 800 simulated cases that had less than 0.5Å difference. The vertical lines represent values of ϵ for cases in Table 1. It appears that most of the ϵ values are between 0.0 and 1.5 to minimize the error in the measurement (Figure 4, right). However, we observed larger ϵ values for the experimentally derived data than for the simulated density maps. This difference is likely to be associated to the quality of the skeleton and helix detection. For the simulated cases, ϵ between 0.0 and 1.5 is more likely to produce a good estimate given good preprocessing of the density maps.

Conclusions

We have developed a new approach to estimate loop length along the skeleton from a CryoEM density map. Our tests, using both simulated and experimentally derived images at medium resolution, show that it is possible for our proposed method to estimate fairly accurately the loop length along the skeleton if the SSEs such as α-helices and the skeleton are detected fairly accurately.

Methods

The overall process to measure the loop length along the skeleton contains two tasks: preprocessing and length calculation (Figure 5). The purpose of the preprocessing is to derive the skeleton and the endpoints of the two helices from the density map. Once such information is obtained, our algorithm uses graphs and computational geometric concepts to derive the simplified curve.

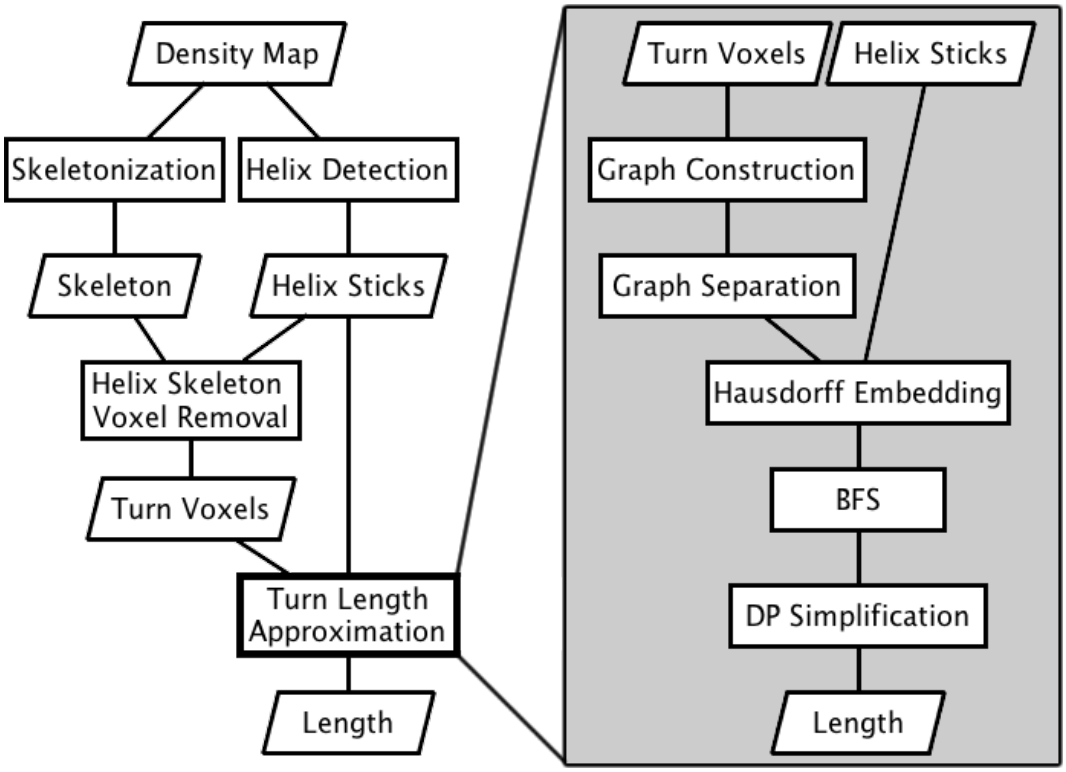


Figure 5: The process of loop length estimation. (Left) preprocessing, and (right) the length estimation algorithm.

Preprocessing

Each case in Table 1 had a density map generated using the HLH segment of the PDB structure and EMAN's *pdb2mrc* [29]. We applied a skeletonization method that utilizes the local maximum points and clustering to derive the skeleton points from the density map. The HLH regions of cases in Table 2 were extracted from entire density images downloaded from EMDB. We used SSETracer, a secondary structure detection method to detect helices from the density map. It is modified from SSELearner [19] with improved speed. Since helix detection is independent from skeletonization, it is necessary to remove the skeleton voxels that belong to the helix region in order to obtain the skeleton belonging to the loop. We removed those skeleton voxels that are within 2.3Å of the central axis of the helix. Note that a helix is 2.3 - 2.5Å in radius [14, 30]. After such processing, the skeleton voxels that presumably belong to the loop are segmented from the rest of the skeleton voxels and are subject for length calculation.

Algorithm

Local connectivity graphs

A local connectivity graph (LCG) represents a cluster of skeleton voxels. We impose a constraint on the maximum allowable edge length in a graph, possibly yielding multiple disconnected graphs when all skeleton voxels are considered. For our tests, we normalized the distances between the image's voxels to unity, and chose a maximum edge length $l < 2$, producing individual connected subcomponents if they can be clustered into distant groups, referred to as LCGs in this paper.

Selecting connected components

Oftentimes, segmented or sparse density data yield multiple LCGs. Also, in general, it is not known which helix endpoints the loop actually lies between. We must then determine the best LCG for each possible pair of helix endpoints. For two helices, one with endpoints p and q and the other with r and s, there exists a set Z of four possible endpoint pairs: $Z := \{\{p, r\}, \{p, s\}, \{q, r\}, \{q, s\}\}$. For each endpoint pair $z \in Z$, let the directed Hausdorff distance to an LCG [31] be defined as

$$h(z, b) = \max_{z_i \in z} \min_{b_j \in b} d(z_i, b_j), \tag{1}$$

where z is the set of helix endpoints (comprised of voxels denoted z_i) and b is an LCG (comprised of voxels denoted b_j) from the set B of all LCGs; $d(z_i, b_j)$ is then the Euclidean distance between a helix endpoint voxel and LCG voxel. In the presence of multiple LCGs, we choose the best LCG $\hat{l}_z$ per endpoint pair $z \in Z$

by taking the minimum directed Hausdorff distance over all LCGs:

$$\hat{l}_z = \min_{b \in B} h(z, b). \tag{2}$$

We can then use the voxels of $\hat{l}_z$ to build our model of the loop between the endpoints of z.

It should be noted here that the directed Hausdorff is not commutative–in general, $h(M, N) \neq h(N, M)$–and we always chose M as a set (pair) of helix endpoints, and N as an LCG. Figure 6 shows the configuration for case 30 (PDB 1O6L) from Table 1, where we want to find $\hat{l}_z$ among the set of LCGs $B := \{1, 2, 3, 4, 5, 6\}$ to search for the loop that *may* lie between the helix endpoint pair a. After finding $\hat{l}_z$ using equation (2), we do the same thing for each other helix endpoint pair. We try connecting the helix endpoints to their respective closest voxels in $\hat{l}_z$ w.r.t. Euclidean distance. If either of the new edges connecting p or r is longer than 5Å, we discard the combination as an infeasible path.

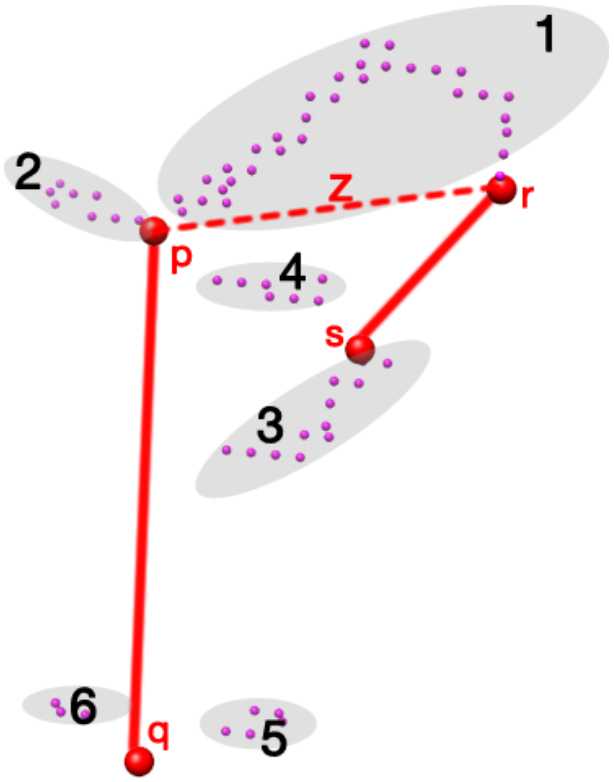


Figure 6: Two detected helices (solid red lines), with a pair z of helix endpoints (connected by the red dashed line) and several LCGs (gray ellipses) from PDB 1O6L. In this case, LCG *1* is closest to z in terms of directed Hausdorff distance.

Pathfinding

After finding the best LCG for a given possible helix endpoint pair, the next step is constructing a path that traverses it in a way that will approximate the loop. We simply performed a breadth-first search starting from one of the helix endpoints we added, and reconstruct the path that ends at the other one in the graph [32], with a helix endpoint as the source. For a given HLH, we find four such paths, one for each possible helix endpoint pair.

Path simplification

Ideally, the distance between two specific ends of two helices should be measured along the skeleton connecting the two ends by using our initial path. If we simply add the length of the line segments along the initial path, there is a danger of over estimation due to the potential zigzagging induced from drawing a path along the edges of the cubic lattice of the 3D image. To overcome this, we simplify the line by systematically removing points using the Douglas-Peucker algorithm [33, 34] generalized to three dimensions by modifying de Halleux's implementation given in [35].

The Douglas-Peucker line simplification algorithm allows us to remove points from a piecewise-linear 3D line (referred to here as a polyline), such that the resulting polyline remains within some tolerance epsilon ϵ from the original one. Consider a two-dimensional example as in Figure 7. Part (i) shows an initial polyline $\overline{a...b}$. The algorithm is recursive, and takes as parameters the tolerance ϵ (Figure 7 (ii)) and a multi-point segment of a polyline. At each recursive iteration it finds an interior point of the current segment which is the most distant from the straight line connecting the end points of the segment, as in Figure 7 (ii) and (iii). If the most distant point is within ϵ from the straight line, the segment is replaced by the straight line, and all interior points are removed. Otherwise, the segment is split into two sub-segments by this most distant interior point, and the algorithm proceeds recursively on each sub-segment. In Figure 7 (iii), $\overline{ac}$ and $\overline{cb}$ are treated in different recursive calls; e is the farthest point from $\overline{cb}$, and no points lie outside the epsilon band for $\overline{ac}$. Overall, the initial polyline $\overline{a...b}$ is simplified into polyline $\overline{aceb}$, which approximates the length of the loop between helix endoints.

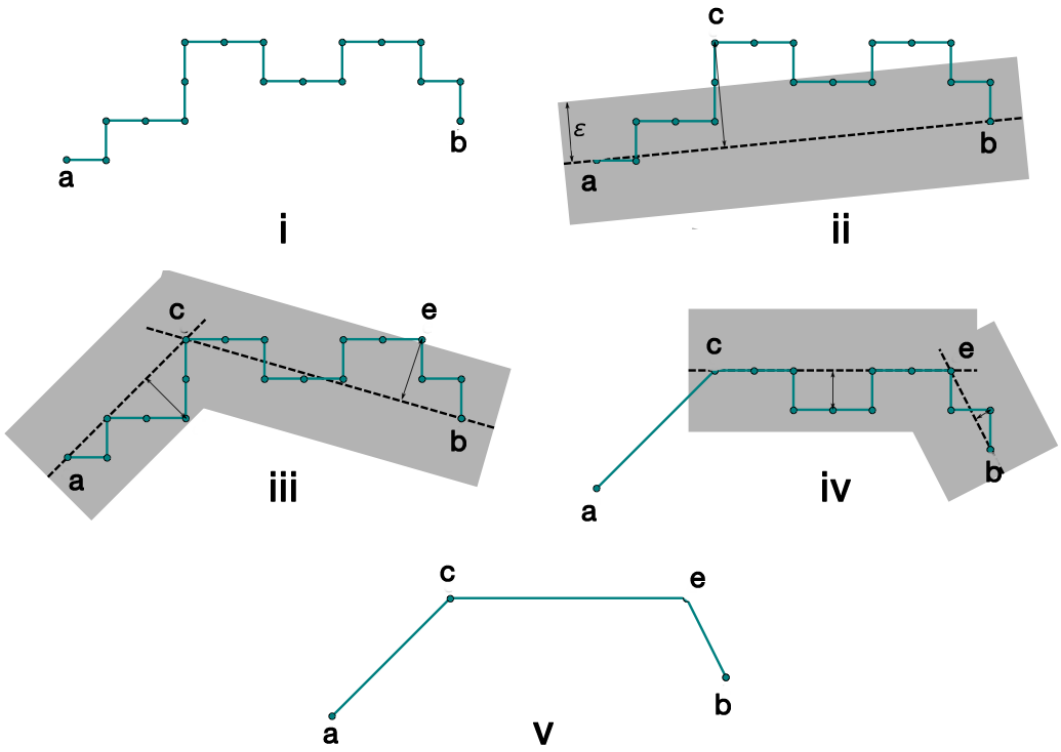


Figure 7: Recursive iterations of the Douglas Peucker line simplification algorithm. Each gray region, as in (ii), illustrates the distance from the test line ($\overline{ab}$ in (ii)) defined by ϵ.

List of abbreviations

CryoEM - cryoelectron microscopy

SSE - secondary structure element - either α-helices or β-sheets

EMDB - Electron Microscopy Data Bank

PDB - Protein Data Bank

HLH - helix-loop-helix motif found in protein structures

LCG - local connectivity graph - a connected graph of skeleton voxels with a maximum allowed edge length

Author's contributions

Andrew McKnight compiled the test set and developed the software for the algorithm. Jing He, as advisor, guided through the problem she has previously researched regarding SSE topology matching. Nikos Chrisochoides and Andrey Chernikov provided technical information and guidance for our algorithm. Dong Si developed SSETracer and Kamal Al Nasr developed our new skeletonizer, and both also provided assistance with background and development.

Acknowledgements

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Table 1 - Accuracy of the loop length estimation in the simulated data set.

ID: PDB ID from which the loop came; AA: the number of amino acids in the loop; *Expected* = $(AA + 1)$ * 3.8Å; *Measured*: the estimated length of the loop along the skeleton or its simplification; *Diff*: *Measured* - *Expected*; *RelErr*: *Difference/Expected*; *DP* ϵ is the value that produced the minimum *Diff* in the estimation.

No	ID	AA	Expected	Measured	Diff	RelErr	DP ϵ
1	1ARO	1	7.6	7.4396	0.1604	2.1	1.00
2	1B0B	1	7.6	7.7384	0.1384	1.8	1.25
3	1BGP	1	7.6	7.6755	0.0755	1.0	1.30
4	1BQB	1	7.6	8.0995	0.4995	6.6	2.30
5	1GUX	1	7.6	7.8102	0.2102	2.8	6.00
6	1B43	2	11.4	11.4264	0.0264	0.2	0.45
7	1B89	2	11.4	11.8811	0.4811	4.2	2.55
8	1BD8	2	11.4	11.3578	0.0422	0.4	0.00
9	1BPY	2	11.4	11.4800	0.0800	0.7	2.25
10	1BR1	2	11.4	11.1461	0.2539	2.2	0.00
11	1FJL	3	15.2	15.4724	0.2724	1.8	1.35
12	1FK5	3	15.2	14.9523	0.2477	1.6	0.00
13	1FUR	3	15.2	15.2643	0.0643	0.4	6.00
14	1H0M	3	15.2	15.3601	0.1601	1.1	2.70
15	1DU0	3	15.2	14.9900	0.2100	1.4	0.60
16	1A87	4	19.0	18.8901	0.1099	0.6	0.95
17	1AIH	4	19.0	19.2057	0.2057	1.1	6.00
18	1AJ8	4	19.0	4.1231	14.8769	78.3	0.00
19	1BMT	4	19.0	19.2313	0.2313	1.2	5.55
20	1BOU	4	19.0	18.9609	0.0391	0.2	0.70
21	1D8L	5	22.8	23.1403	0.3403	1.5	0.60
22	1DI1	5	22.8	22.9243	0.1243	0.5	4.25
23	1DLC	5	22.8	22.5618	0.2382	1.0	0.00
24	1DNP	5	22.8	23.1044	0.3044	1.3	1.70
25	1DP7	5	22.8	22.7786	0.0214	0.1	2.10
26	1CQX	6	26.6	26.2583	0.3417	1.3	0.00
27	1CSH	6	26.6	26.9157	0.3157	1.2	1.85
28	1HM6	6	26.6	7.1461	18.8539	26.3	0.00
29	1MW8	6	26.6	26.2419	0.3581	1.3	0.00
30	1O6L	6	26.6	26.6271	0.0271	0.1	6.00
31	1DJX	7	30.4	30.7842	0.3842	1.3	3.85
32	1E5Q	7	30.4	30.5342	0.1342	0.4	4.65
33	1FFV	7	30.4	30.0703	0.3297	1.1	2.50
34	1H99	7	30.4	30.1897	0.2103	0.7	0.00
35	1IRX	7	30.4	30.7213	0.3213	1.1	6.00
36	1O6L	8	34.2	34.6762	0.4762	1.4	6.00
37	1QVR	8	34.2	34.2838	0.0838	0.2	0.60
38	1S0V	8	34.2	34.2505	0.0505	0.1	0.95
39	1TAU	8	34.2	34.3267	0.1267	0.4	0.70
40	1U09	8	34.2	34.1468	0.0532	0.2	2.05
41	1D6M	9	38.0	38.1574	0.1574	0.4	1.00
42	1FUR	9	38.0	38.3249	0.3249	0.9	2.85
43	1H32	9	38.0	38.1491	0.1491	0.4	0.70
44	1QPC	9	38.0	37.9111	0.0889	0.2	0.00
45	1SU8	9	38.0	37.9337	0.0663	0.2	0.65
46	1QRT	10	41.8	41.7369	0.0631	0.2	0.75
47	1R1H	10	41.8	41.3131	0.4869	1.2	0.00
48	1RJB	10	41.8	41.8528	0.0528	0.1	1.00
49	1XO0	10	41.8	41.8814	0.0814	0.2	1.05
50	2B63	10	41.8	41.4589	0.3411	0.8	4.60

Table 2 - Accuracy of the measured loop length for the experimentally derived CryoEM data.

See Table 1 for notations except *ID*: the EMDB ID in which the loop was tested.

No	ID	AA	Expected	Measured	Diff	RelErr	DP ϵ
1	5030	1	7.6	9.5128	1.9128	25.2	6.00
2	5138	1	7.6	8.2690	0.6690	8.8	6.00
3	5138	2	11.4	11.5490	0.1490	1.3	2.35
4	1733	3	15.2	14.3661	0.8339	5.5	4.05
5	1733	3	15.2	15.0790	0.1210	0.8	3.80
6	5001	3	15.2	11.1189	4.0811	26.8	0.00
7	5001	3	15.2	12.5132	2.6868	17.7	0.00
8	5001	3	15.2	15.6095	0.4095	2.7	2.35
9	5030	3	15.2	15.3747	0.1747	1.1	6.00
10	5030	3	15.2	14.6116	0.5884	3.9	1.75
11	5030	3	15.2	15.1321	0.0679	0.4	3.50
12	5138	3	15.2	14.2916	0.9084	6.0	5.30
13	1733	4	19.0	18.2477	0.7523	4.0	0.00
14	5001	4	19.0	19.1872	0.1872	1.0	6.00
15	5168	4	19.0	21.8790	2.8790	15.2	6.00
16	1740	5	22.8	26.4127	3.6127	15.8	6.00
17	1740	6	26.6	29.3993	2.7993	10.5	6.00
18	5168	6	26.6	22.4231	4.1769	15.7	0.00